

Final Report Grading Rubric

CS 789 - Software Reengineering

Milestone 3 | 100 points (25 team + 75 individual)

Team		Student	
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The Final Report is the consolidated account of the project. Each student is assessed on all 7 assigned PRs and both LLM-assisted activities; the team is assessed on synthesis, coordination, tool/pattern evaluation, and the coherence of the final account.

Team Component [25 points]

Criterion	Pts	Full-credit evidence	Score
Response to team-level feedback	2.0	Team-level feedback from the Intermediate Report and Nov. 16 session is traceably addressed; unresolved earlier items are carried through. Individual feedback is assessed in the student component.	
Final project context, strategy, and outcome summary	2.0	Final scope, workflow, Definition of Done, patterns/maxims, PR allocation, and outcomes are concise, current, and free of obsolete planning language.	
Project coordination and delivery	3.0	Weekly meetings/minutes, Discord, progress tracking, blocker escalation, and individual ownership show sustained coordination through project completion.	
Cross-PR analysis	5.0	Team synthesizes evidence across categories to explain patterns in clean integrations, failures, divergence, refactorings/design evolution, testing challenges, and RePatch capabilities/limits.	
Team-level LLM synthesis	3.0	Team compares individual Part A/Part B results, identifies recurring successes/failures and usable-test patterns, and explains where developer judgement remained necessary.	
Evaluation of tools and techniques	3.0	Team evaluates the techniques actually used across the five lab areas, explaining what each revealed/accomplished and its limitations.	
Reengineering patterns and project-wide lessons	3.0	Team connects OORP patterns to concrete evidence across conventional and LLM-assisted work and derives lessons that generalize beyond individual PRs.	
Reflection, conclusion, coherence, and supporting evidence	4.0	Team reflection, conclusion, individual-contribution mapping, references, and repository/minutes links form a coherent, traceable final account without unnecessary raw logs.	
Team Component subtotal	25		

Individual Component [75 points]

Criterion	Pts	Full-credit evidence	Score
Response to individual feedback, if applicable	2.0	Student resolves feedback directed to their own work and makes the revision traceable. If no individual feedback was given, award these points based on appropriate revision and carry-forward of the student's prior work.	
Completion and traceability of all 7 PRs	5.0	All seven assigned PRs are complete/documented, clearly attributed, and consistent with the final PR summary and repository evidence.	
Repository investigation and change understanding	6.0	Student accurately explains the purpose, provenance, changed files/components, and relevant history for assigned PRs and uses that evidence in the analysis.	
Design recovery, design gap, and adaptation/barrier analysis	13.0	For each PR, student recovers relevant source/target design, explains the design/evolution gap, and shows resulting target design/adaptation where appropriate; Category 4 focuses on defensible design/refactoring barriers.	
Integration, RePatch, refactoring, and manual/failure analysis	13.0	Integration outcomes are technically sound and evidence-based; RePatch/refactorings are analyzed where applicable; manual adaptation is targeted; Category 4 is diagnosed without forcing redesign.	
Testing, coverage/test adequacy, and validation	11.0	Student runs and interprets relevant tests, assesses adequacy, adds/adapts tests where needed, uses comparable coverage where valid, and explains limitations rather than relying on percentages alone.	
LLM-assisted Part A - patch integration	8.0	Student performs the Part A experiment on an assigned Category 4 PR with baseline failure, deliberate context selection, reproducible prompts/iterations, verification, pattern-based reasoning, and a defensible outcome. Success is not required.	
LLM-assisted Part B - regression check and test generation	8.0	Student performs Part B on an eligible integrated PR, establishes regression/test baseline, validates generated tests including can-it-fail reasoning, reports generated vs usable tests, and uses coverage/other adequacy evidence appropriately.	
Reengineering reasoning and OORP patterns	6.0	Student connects patterns to concrete decisions across PR and LLM-assisted work and articulates lessons rather than merely naming patterns after the fact.	
Quality and reproducibility of individual evidence	3.0	Individual evidence is concise, attributable, reproducible, and consistent across report and repository artifacts; model assistance is clearly disclosed.	
Individual Component subtotal	75		

Scoring guidance

Full credit requires technically sound, traceable, and appropriately explained evidence. Award partial credit in proportion to completeness and quality. Team points are shared; individual points are assigned separately. A technically unsuccessful integration or LLM experiment can earn full credit when the required process, verification, evidence, and analysis are complete and well reasoned.

Component	Maximum	Score
Team	25	
Individual	75	
Total	100	

Grader comments: